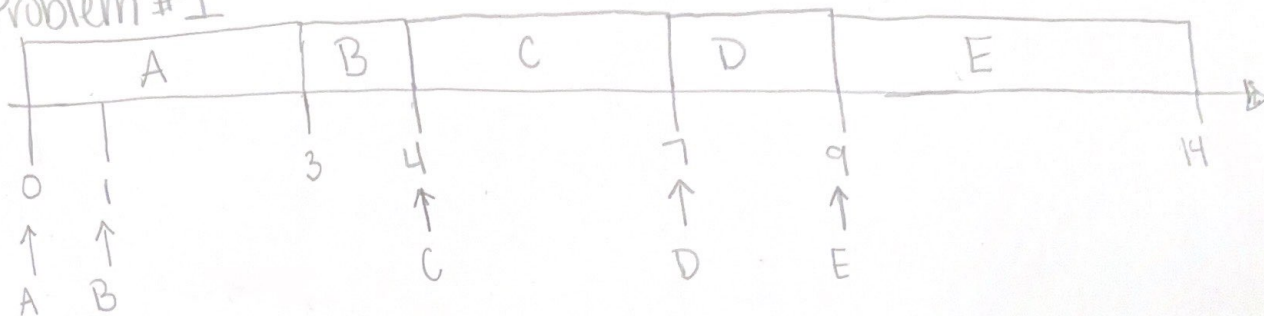


Written Assignment #4

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Problem #1

(A)



Completion time

$$A = 3$$

$$C = 7$$

$$E = 14$$

$$B = 4$$

$$D = 9$$

FCFS

Avg. Turnaround

$$A = 3 - 0 = 3$$

$$B = 4 - 1 = 3$$

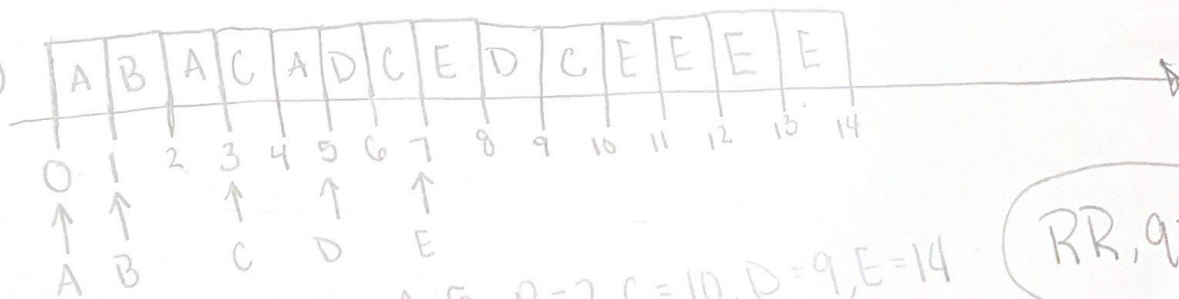
$$C = 7 - 3 = 4$$

$$D = 9 - 4 = 5$$

$$E = 14 - 5 = 9$$

$$\frac{3+3+4+5+9}{5} = 4.8$$

(B)



Completion time: $A=5, B=2, C=10, D=9, E=14$

RR, $q=1$

Avg. Turnaround

$$A = 5 - 0 = 5$$

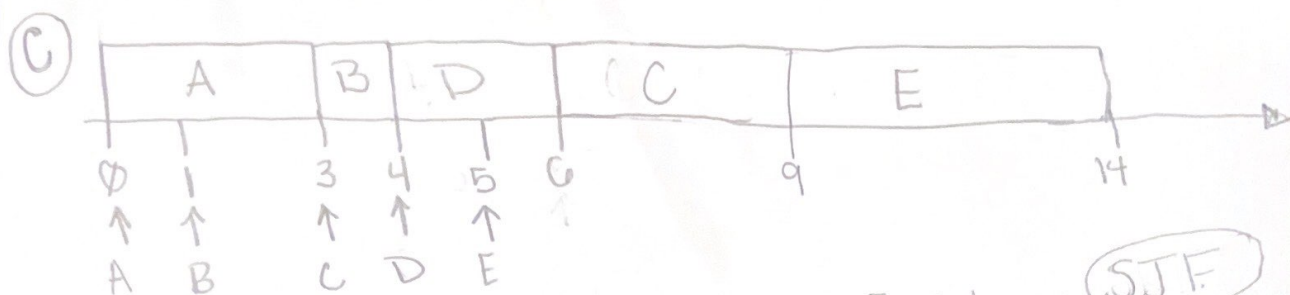
$$B = 2 - 1 = 1$$

$$C = 10 - 3 = 7$$

$$D = 9 - 5 = 4$$

$$E = 14 - 7 = 7$$

$$\frac{5+1+7+4+7}{5} = 4.8$$



(SJF)

Completion time: $A=3, B=4, C=9, D=6, E=14$

Avg. Turnaround:

$A: 3-0=3$

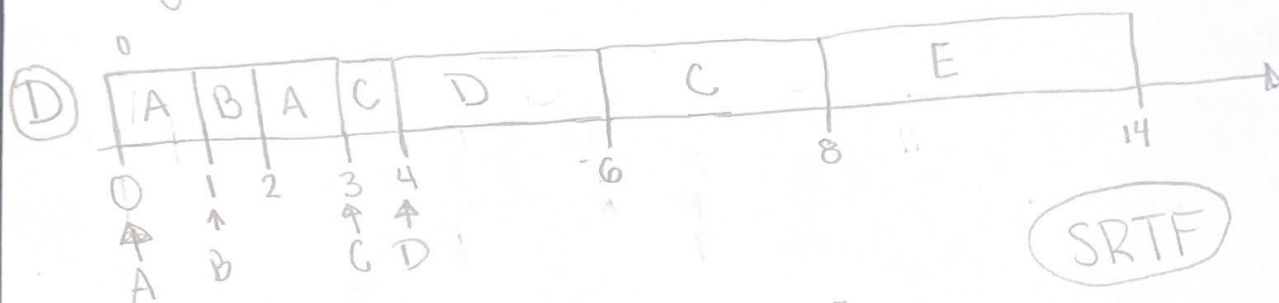
$B: 4-1=3$

$C: 9-3=6$

$D: 6-4=2$

$E: 14-5=9$

$$\frac{3+3+6+2+9}{5} = \frac{23}{5} = 4.6$$



(SRTF)

completion time $A=3, B=2, C=8, D=6, E=14$

Avg Turnaround

$A: 3-0=3$

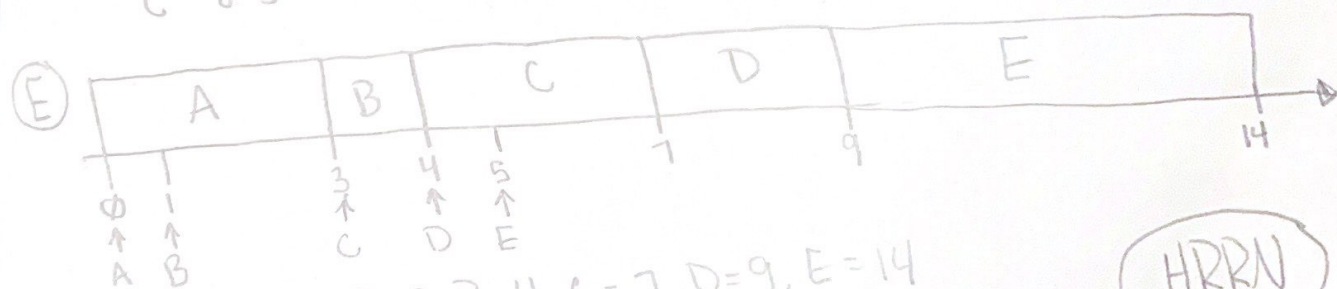
$B: 2-1=1$

$C: 8-3=5$

$D: 6-4=2$

$E: 14-5=9$

$$\frac{3+1+5+2+9}{5} = 4$$



(HRRN)

completion time: $A=3, B=4, C=7, D=9, E=14$

Average Turnaround:

$A: 3-0=3$

$B: 4-1=3$

$C: 7-3=4$

$D: 9-4=5$

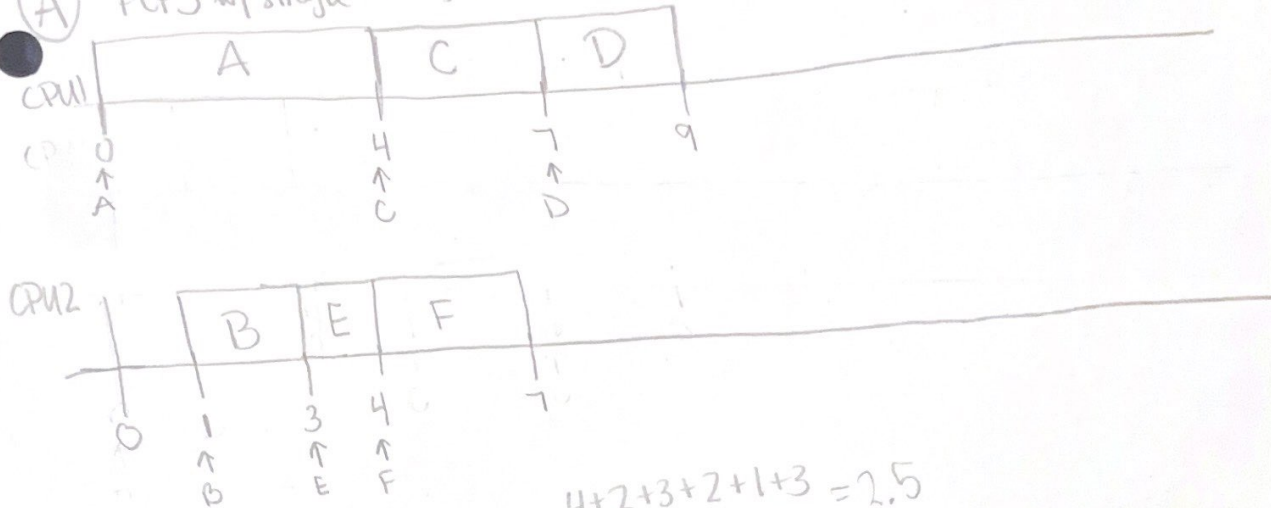
$E: 14-5=9$

$$\frac{3+3+4+5+9}{5} = 4.9$$

Problem #2

Felicity Lester

(A) FCFS w/ single ready queue. when both see pick 1



$$\frac{4+2+3+2+1+3}{6} = 2.5$$

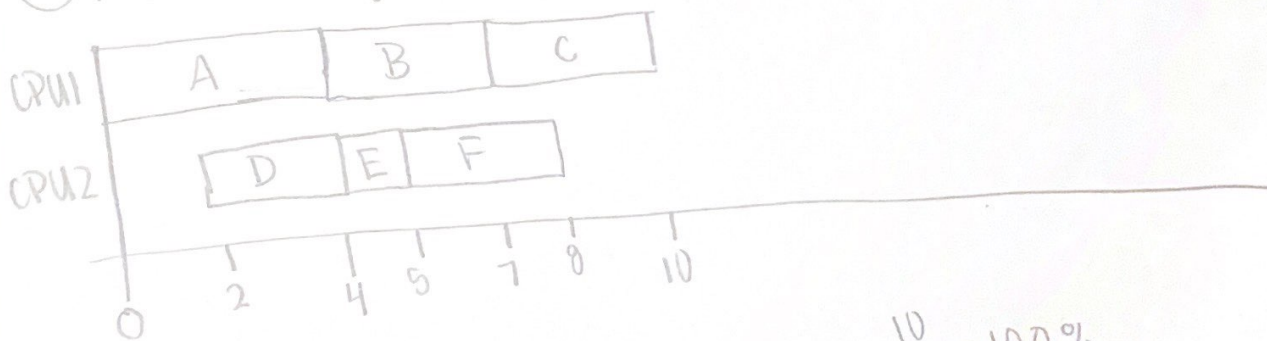
Turnaround Time

$$\begin{array}{ll} A & 4-0=4 \\ B & 3-1=2 \\ C & 7-4=3 \end{array} \quad \begin{array}{ll} D & 7-9=2 \\ E & 4-3=1 \\ F & 7-4=3 \end{array}$$

$$CPU1 = \frac{9}{9} = 100\%$$

$$CPU2 = \frac{6}{7} \approx 86\%$$

(B) FCFS w/ Ready Queue per CPU



Turnaround

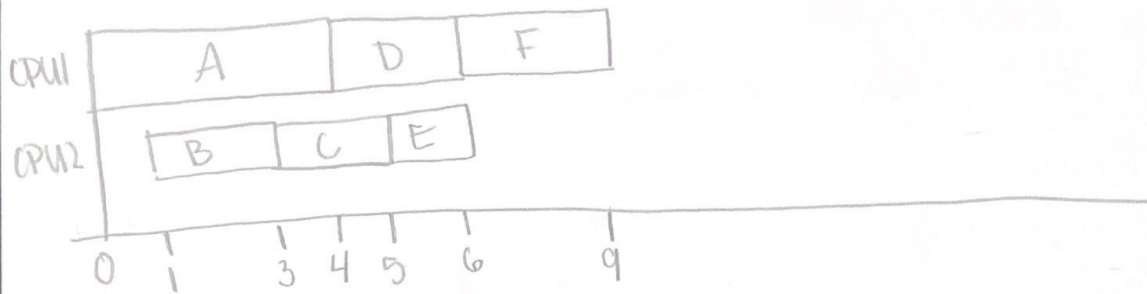
$$\begin{array}{ll} A & 4-0=4 \\ B & 7-1=6 \\ C & 10-2=8 \\ D & 4-2=2 \\ E & 5-4=1 \\ F & 7-5=2 \end{array}$$

$$\frac{4+6+8+2+1+2}{6} = 4$$

$$CPU1 = \frac{10}{10} = 100\%$$

$$CPU2 = \frac{6}{10} = 60\%$$

© HRRN w/ single ready queue



Turnaround

$$A \ 4 - 0 = 4$$

$$B \ 3 - 1 = 2$$

$$C \ 5 - 3 = 2$$

$$D \ 6 - 4 = 2$$

$$E \ 6 - 5 = 1$$

$$F \ 9 - 6 = 3$$

$$\frac{4 + 2 + 2 + 2 + 1 + 3}{6} \approx 2.333$$

$$CPU1 = \frac{9}{9} = 100\%$$

$$CPU2 = \frac{5}{6} \approx 83\%$$

Problem #3

Felicity

- $t_1 (3,1) R$
- $t_2 (5,2) B$
- $t_3 (8,3) G$

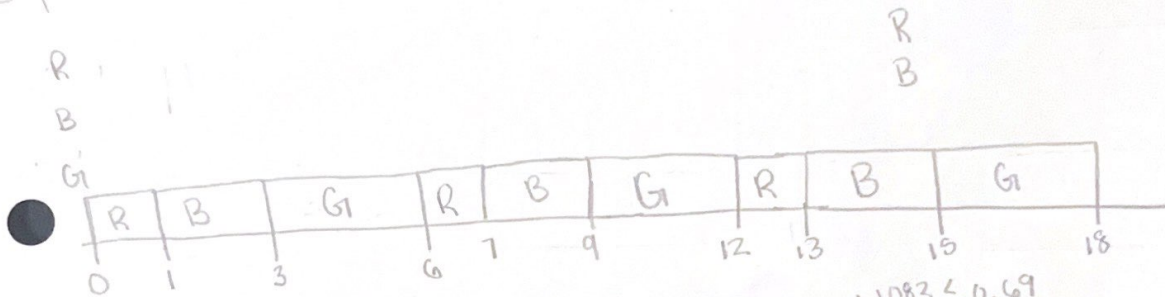
(A) Total Utilization

$$\frac{1}{3} + \frac{2}{5} + \frac{3}{8} = 1.1083$$

$1.1083 \leq 1.00$ misses possible

$$EDF \leq 1.00$$

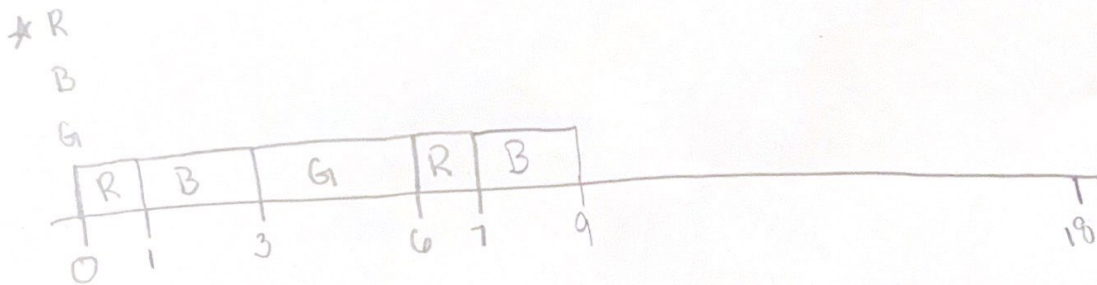
(B) preemptive EDF in interval $(0,18)$



$$1.1083 \leq 0.69$$

$U \leq 0.69$ misses

(C) preemptive RM w/ interval $(0,18)$



(D) reduce execution time w/ period 3

$$\frac{3}{8} = 0.375$$

$$1 - 0.375 = 0.625 \text{ min}$$